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Unit – 2 : Session – 3 : SLO – 1 | Advantages and Disadvantages of an Array

■ I. Advantages — Fill in the Blanks

1. direct (random/constant-time) — index-based $O(1)$ access.
2. contiguous — elements sit in contiguous memory locations.
3. fixed (large) — good for storing a fixed number of similar-typed items.
4. fixed — size is fixed once declared.
5. repetitive (iterative) — loops make bulk operations easy.
6. overhead (need) — no need to declare many separate variables.
7. known (exact/sufficient) — memory is efficient when the size matches the need.
8. constant ($O(1)$) — index lookup time.

■ II. Disadvantages — Fill in the Blanks

1. compile (declaration) — size must be known at compile time (for static arrays).
2. fixed — size cannot change at runtime.
3. shifting — middle insertion requires shifting other elements.
4. shifting — deletion also involves shifting elements.
5. too large (over-declared) — over-declaring wastes memory.
6. same — only one data type per array.
7. $O(n)$ (linear) — worst-case search time in an unsorted array.